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Laurea Magistrale in ‘Robotics And Automation Engineering’

Corso: Optimal Control

Model and Control of A Hot Air Balloon By LQ Strategy

Studente

Alessandro Furina
252328

Professore

Domenico Famularo

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INTRODUCTION

The aim of this project consists of stabilizing and designing a controller for a hot air balloon. Stabilization and control are common problems in the automatic framework, because often there is the need of satisfying functional requirements in terms of performance. In the set of all the possible solutions in charge to guarantee stability and control, we are interested in the LQ technique, that is to say Linear Quadratic Strategy.

1. MODELING

As introduced previously, the system to analyse is a hot air ballon. In order to model the dynamic, fundamental step for our purposes, we need to identify some variables of interest:

Fig. 1.1: Definition of variables z and ż

As shown in *Fig. 1.1*, first we can identify two important variables which are: the z coordinate of the balloon representing its altitude with respect to the ground (origin of the axle) and the evolution of the z coordinate, that is to say the z-velocity (negative if the ballon decreases its altitude, positive vice versa). They are expressed respectively in [*m*] and [*m/s*].

Moreover, we need information about the evolution of the air temperature inside the envelope. It's known that the flight of balloons is guaranteed by the increasement of the air temperature inside the envelope. The heating of the air, managed by the burner, generates a difference of density between the envelope of the balloon and the environment around it. Because of hot air has lower density than fresh air, convective forces arise allowing the balloon to take off. Thus, we can introduce another variable:

Fig.1.2: Definition of θ variable

It is expressed in Kelvin and gives information over time on the temperature inside the envelope.

The last variable to consider, it is easy to imagine, is represented by the power of the burner:

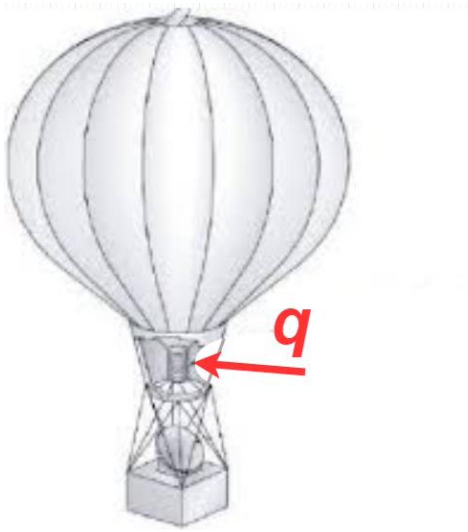

Fig 1.3: Definition of q variable

It is expressed in Watt and gives information over time on the power flow inside the envelope. In can be assumed as the input of the system because it allows to change the state of the balloon.

Now, the identification of all the variables of interest is completed. Next step consists of obtaining a I-S-U local representation for the system. The following model can be considered as a good approximation of the evolution of these variables:

$$\begin{cases} \ddot{z} = -\frac{b}{m}\dot{z} - g + \frac{V}{m} g \rho_a \left(1 - \frac{\theta_a}{\theta}\right) \\ \dot{\theta} = -\frac{k}{c}\theta + \frac{k}{c}\theta_a + \frac{q}{c} \end{cases} \qquad \textbf{\textit{(1.1)}}$$

Where:

- **m** represents the mass of the balloon: 800 [kg].
- **b** is the air drag coefficient: 20 [N*s/m].
- **g** is the gravity acceleration: 9.81 [m/s^2].
- **V** is the volume of the envelope: 2000 [m^3].
- $\boldsymbol{\rho_a}$ represents the air density: 1.184 [Kg/m^3].
- $\boldsymbol{\theta_a}$ is the air temperature around the balloon: 298 [K].
- $\boldsymbol{c}$ represents the heat capacity of the air in the envelope: 718 [J/K].
- $\boldsymbol{k}$ is the heat transfer coefficient between balloon and environment: 2.5 [W/K].

In order to obtain the I-S-U model, we have to identify: state, input and output. They can be considered as follows:

$$q = u \ \in R,$$

$$\begin{bmatrix} z \\ \dot{z} \\ \theta \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \ \in R^3,$$

$$z = y \ \ \in R$$

Hence, we can replace the old variables with the state-space variables and define the following model for the balloon:

$$\begin{cases} \frac{dx_1}{dt} = x_2 \\ \frac{dx_2}{dt} = -\frac{b}{m}x_2 - g + \frac{V}{m}g\rho_a(1 - \frac{\theta_a}{x_3}) \\ \frac{dx_3}{dt} = \frac{k}{c}\theta_a - \frac{k}{c}x_3 + \frac{u}{c} \\ y = x_1 \\ x(0) = x_0 \end{cases} \qquad \textbf{\textit{(1.2)}}$$

The model is non-linear for the presence of the state x_3 at the denominator in the second differential equation.

2. EQUILIBRIA POINTS

This section is dedicated to the search of the equilibria points. By definition, an equilibrium is a condition in which the system doesn't change its state. This means that there is no variation of the state components and, therefore, of the output. We have already introduced the state-space representation for the system. Looking at that model in a general way we can describe the evolution as follows:

$$\begin{cases} \frac{dx}{dt} = f\big(t, x(t), u(t)\big) \\ y(t) = g\big(t, x(t), u(t)\big) \\ x(0) = x_0 \end{cases} \qquad \textbf{(2.1)}$$

With: $x \in R^3, u \in R, y \in R$

When the system reaches the equilibrium, the evolution becomes:

$$\begin{cases} \frac{dx}{dt} = f\big(t, x_{eq}, u_{eq}\big) \\ y(t) = g\big(t, x_{eq}, u_{eq}\big) \\ x(0) = x_0 \end{cases} \qquad \textbf{(2.2)}$$

Where u_{eq} represents the input that has conditioned the system to reach the equilibrium.

Considering the definition introduced previously, we obtain:

$$\begin{cases} \frac{dx}{dt} = f\big(t, x_{eq}, u_{eq}\big) = 0 \\ y(t) = y_{eq} \\ x(0) = x_0 \end{cases} \qquad \textbf{(2.3)}$$

Thus, the system has reached a steady-state condition.

The search of the equilibria points is fundamental to analyse properties regarding, for example, the stability of these points (capability of the system to maintain the equilibrium whereas internal perturbations are acting) and, possibly, to design a controller.

In order to find the equilibria, let's apply the mathematical definition to the system of differential equations:

$$\begin{cases} f_1\big(x_{eq}, u_{eq}\big) = 0 \\ f_2\big(x_{eq}, u_{eq}\big) = 0 \\ f_3\big(x_{eq}, u_{eq}\big) = 0 \end{cases} \Rightarrow \begin{cases} x_{2eq} = 0 \\ -\frac{b}{m}x_{2eq} - g + \frac{V}{m}g\rho_a(1 - \frac{\theta_a}{x_{3eq}}) = 0 \\ \frac{k}{c}\theta_a - \frac{k}{c}x_{3eq} + \frac{u_{eq}}{c} = 0 \end{cases}$$

Solving this system we obtain the characterization of the equilibria points:

$$\begin{cases} x_{1eq} = whatever\ but\ constant \\ x_{2eq} = 0 \\ x_{3eq} = \frac{V\rho_a\theta_a}{V\rho_a - m} \\ u_{eq} = k\left(x_{3eq} - \theta_a\right) \end{cases} \qquad \textbf{(2.4)}$$

So, to make it possible the balloon remains in the air without increasing or decreasing its altitude ($x_{1eq} = const.$ and $x_{2eq} = 0$) it is sufficient that both x_{3eq} and u_{eq} have their respective values of equilibrium whatever is the altitude.

3. LINEARIZATION

Linearization is a technique that allows to approximate the behaviour of Non-Linear functions to Linear functions around the point on which the linearization is operated. This concept can be also used in the context of dynamical systems. In particular, given the general state-space representation of non-linear system:

$$\begin{cases} \frac{dx}{dt} = f\left(t, x(t), u(t)\right) \\ y(t) = g\left(t, x(t), u(t)\right) \\ x(0) = x_0 \end{cases}$$

The approximated linear system can be obtained as follows:

$$\Sigma_{linearized} \begin{cases} \frac{d}{dt}\tilde{x}(t) = \frac{\partial f}{\partial x}\left(x_{eq}, u_{eq}\right) * \tilde{x}(t) + \frac{\partial f}{\partial u}\left(x_{eq}, u_{eq}\right) * \tilde{u}(t) \\ \tilde{y}(t) = \frac{\partial g}{\partial x}\left(x_{eq}, u_{eq}\right) * \tilde{x}(t) + \frac{\partial g}{\partial u}\left(x_{eq}, u_{eq}\right) * \tilde{u}(t) \\ \tilde{x}(0) = \tilde{x}_0 \end{cases} \qquad \textbf{(3.1)}$$

Where:

- The couple $\left(x_{eq}, u_{eq}\right)$ represents the operating point of the linearization. In this case it has been chosen as a point of equilibrium for next purposes about the control.
- The variables $\tilde{x}(t)$, $\tilde{y}(t)$, $\tilde{u}(t)$ are the new state-space variables and they represent the errors between the perturbated solutions with respect to the equilibrium:

  $$\tilde{x}(t) = x(t) - x_e, \tilde{y}(t) = y(t) - y_e, \tilde{u}(t) = u(t) - u_e, \tilde{x}_0 = x_0 - x_e$$

  With $x(t)$,$y(t)$,$u(t)$ obtained as perturbation of the equilibrium solution with small quantities.
- $\frac{\partial f}{\partial x}\left(x_{eq}, u_{eq}\right) = A$, $\frac{\partial f}{\partial u}\left(x_{eq}, u_{eq}\right) = B$, $\frac{\partial g}{\partial x}\left(x_{eq}, u_{eq}\right) = C$, $\frac{\partial g}{\partial u}\left(x_{eq}, u_{eq}\right) = D$

Applying this technique to the Balloon system (Non-Linear) we end up with its linear version:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{b}{m} & \frac{V}{m} g \rho_a \theta_a \frac{1}{{x_{3eq}}^2} \\ 0 & 0 & -\frac{k}{c} \end{bmatrix} \qquad A \in\ R^{3x3},$$

$$B = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{c} \end{bmatrix} \qquad B \in R^{3x1}\ , \qquad \textbf{\textit{(3.2)}}$$

$$C = [1 \quad 0 \quad 0] \qquad C \in R^{1x3}\ ,$$

$$D = 0 \qquad D \in R$$

4. LQ REGULATION

Now there are all the information to go deeper into the core of this project. As introduced at the beginning, the task of this work consists of stabilizing the system and designing a controller in charge of guaranteeing some signals to be tracked. In this section we put our attention to the first problem, the stabilization one, which is the analysis and possibly the stabilization of a point of equilibrium. Thus, we are referring to the internal stability that is the capability of the system to maintain the equilibrium whereas internal perturbations are acting (locally asymptotic stability property). Because of we are facing with a non-linear system, in the next steps we can't infer about the stability of the entire system. Equilibria points of Non-linear systems, in fact, do not share this property, each point can be stable or unstable. However, in this particular case, the dynamic matrix A of the linearized system does not depend on the equilibrium chosen. There is only a dependence on x_{3eq}, but this quantity as we said in the Equilibria Points section is fixed. So, the eigenvalues of the matrix A are fixed too and equal to:

$$\lambda_1 = 0, \lambda_2 = -0.025, \lambda_3 = -0.0035$$

Applying the *Reduced Lyapunov Criterium* we can state that all the equilibria points of the system are unstable for the presence of λ_1.

The task consists of choosing a point of equilibrium and stabilize it. There are many techniques to solve this problem, but we want to focus on the LQ strategy.

An LQ problem is an optimization problem in which it is required to reduce the following quadratic Performance Index:

$$\min u_k \sum_{k=0}^{N-1} ({x_k}^T Q x_k + {u_k}^T R u_k + 2{u_k}^T M x_k) + {x_N}^T S x_N \qquad \textbf{\textit{(4.1)}}$$

Where:

- ${x_k}^T Q x_k + {u_k}^T R u_k + 2{u_k}^T M x_k$ represents the *loss function* with:
  - ${x_k}^T Q x_k \quad , Q = Q^T \geq 0 \ (positive\ semidefinite) \in R^{3x3}$ is an indicator of the distance of the state with respect to the equilibrium during the transient.
  - ${u_k}^T R u_k \quad , \in R$ is an indicator of the energy of the actuator (the burner).
  - $2{u_k}^T M x_k \ , \in R^{1x3}$ is a mixed term.
- ${x_N}^T S x_N \quad , S = S^T \geq 0 \ (positive\ semidefinite) \in R^{3x3}$ represents the *penalty function* and is an indicator of the distance of the final state with respect to the equilibrium.
- N is the number of steps.

The adjective Linear is related to the discrete-time system involved in the optimization:

$$\begin{cases} x_{k+1} = Ax_k + Bu_k \\ y_k = Cx_k + Du_k \\ x_0 = x0 \end{cases} \qquad \textbf{\textit{(4.2)}}$$

The solution of an LQ problem typically consists of reaching a trade-off between the accuracy of the stabilization (how close the final state is to the target) and the consumption of the actuator energy. The aim is to find out the best input signal capable to minimize the performance index ensuring, therefore, the stabilization of the system to the equilibrium point O_x. Any other equilibrium can be reached by adding the bias.

Thus, let's define the model of the system. First, we have to store the parameters:

```
m=800; %[Kg] b=20; %[N*s/m] g=9.81; %[m/s^2] V=2000; %[m^3] rhoa=1.184; %[Kg/m^3]
theta_a=298; %[K] c=718; %[J/K] k=2.5; %[W/K]
parameters=[m,b,g,V,rhoa,theta_a,c,k];
```

The model is implemented in the Simulink file ***Balloon_model.slx*** and here a snapshot of it:

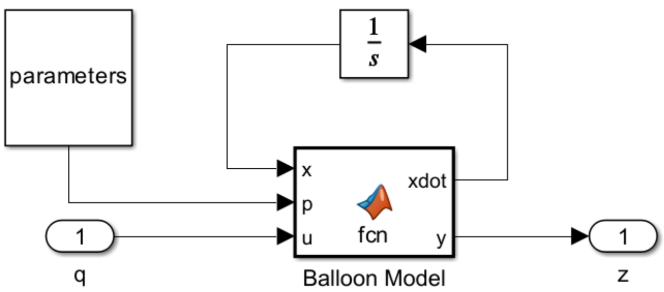

Fig. 4.1: Balloon Model (Simulink)

Inside the matlab function there is the code responsible for the evolution of the system coherently with the equations described in the modeling section.

Now we can try to regulate the system. This operation is implemented in the Matlab file ***Balloon_LQ_regulation.mlx*** and the most important code lines are here reported:

Define the equilibrium:

```
x1eq=900; %personal choice
x2eq=0; %trivial
x3eq=(V*rhoa*theta_a)/((V*rhoa)-m); %according to the equation of the equilibrium
for x3
xeq=[x1eq;x2eq;x3eq];
ueq=k*(x3eq-theta_a); %according to the equation of the equilibrium for u
```

Compute the linearized model around [xeq,ueq]:

```
[Ac,Bc,Cc,Dc]=linmod("Balloon_model",xeq,ueq);
```

Discretize the linearized model in order to apply the LQ strategy:

```
Balloon_linearized_Tc=ss(Ac,Bc,Cc,Dc);
Ts=1; %sampling Time
Balloon_linearized_Td=c2d(Balloon_linearized_Tc,Ts);
[Ad,Bd,Cd,Dd]=ssdata(Balloon_linearized_Td);
```

Define the parameters for the LQ strategy:

```
Cz=[1 0 0; %we want to optimize the altitude (first state component)
    0 0 0];
rho=0.003;
Dzu=[0;
    rho]; %we have just 1 actuator (the burner of the Balloon)
Q=Cz'*Cz;
R=Dzu'*Dzu;
M=Dzu'*Cz;
```

Compute the optimal gains:

```
[F_inf,P_inf,eig_inf]=lqr(Balloon_linearized_Td,Q,R,M');
```

Define the starting point:

```
deltax=[-3;0.2;0.5]; %perturbation of the initial conditions
x0perturbed=xeq+deltax;
```

Some topics about it:

1. The sampling time T_s has been chosen respecting the following rule of thumb:

$$T_s \leq \frac{2\pi}{20\omega_{bw}}$$

With ω_{bw} bandwith pulsation equal to the absolute value of the dominant pole of the linearized matrix A_c. This means that the flight of the Balloon is regulated each second.

2. The parameters for the LQ strategy have been chosen according to the representation of the loss function (without mixed term) by means of the performance output.

$$z_k = C_z x_k + D_{zu} u_k \Rightarrow {x_k}^T Q x_k + {u_k}^T R u_k = {z_k}^T z_k$$

With $Q = {C_z}^T C_z \quad R = {D_{zu}}^T D_{zu}$

3. The optimal gains for the state feedback regulation have been obtained solving the Riccati Backward equation by the *lqr* matlab function.

So far, we have solved the optimization problem which answers to the question How to choose the gain for the state feedback? This operation is an off-line task in the sense that the system is not affected. Now, to regulate the real system (on-line task) it's necessary to project a control loop involving the actuator. This operation is implemented in the Simulink file ***Balloon_LQ_regulation.slx***. here reported:

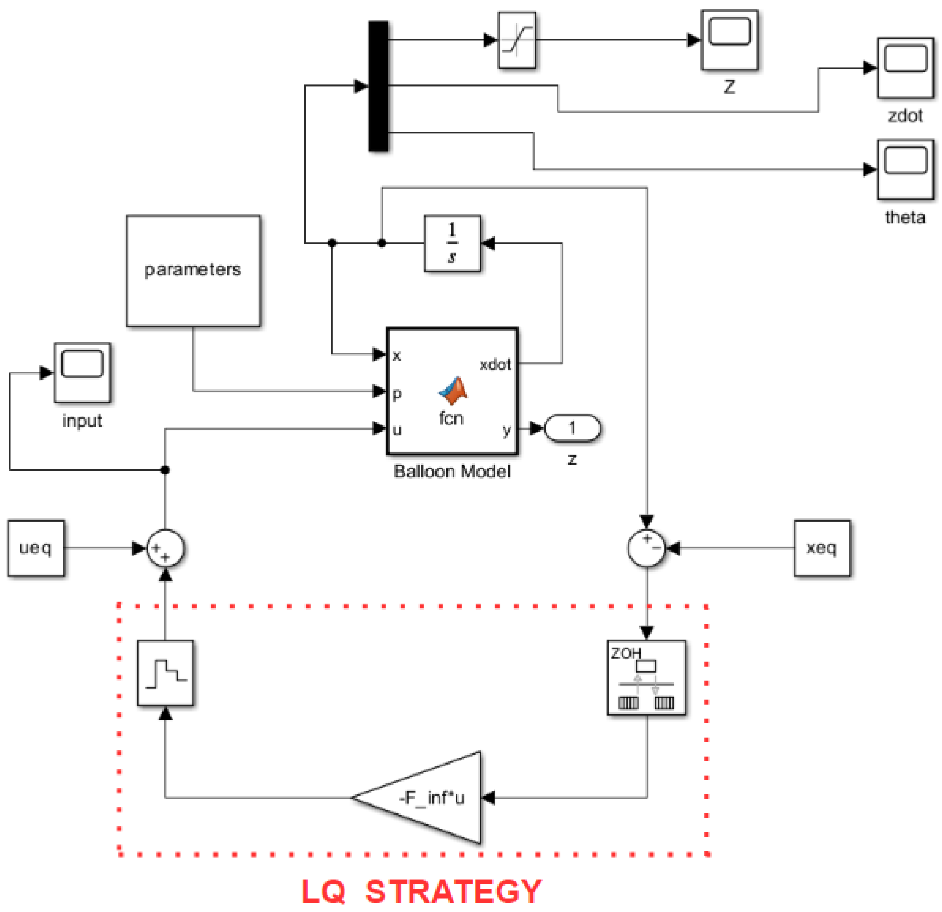

Fig. 4.2: LQ Regulation (Simulink)

First, we remove the bias in order to apply the linear strategy. Secondary, the unbiased state is sampled and the input signal is reconstructed by a Zero Order Holder. The final signal for the actuator is obtained adding the bias related to the input. In order to make impossible a negative altitude (it has no sense in this context) a saturation block is used for the z state.

The results of the simulation are shown in the following figures:

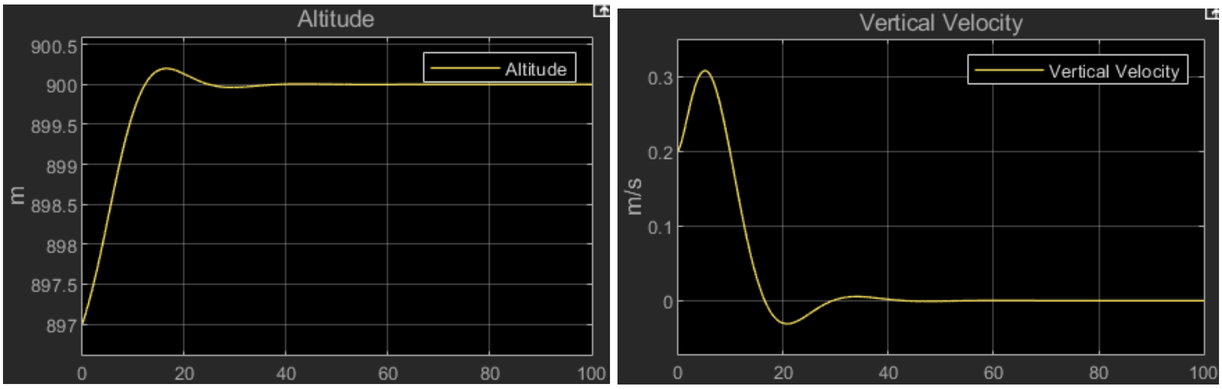

Fig. 4.3: Altitude and Vertical Velocity

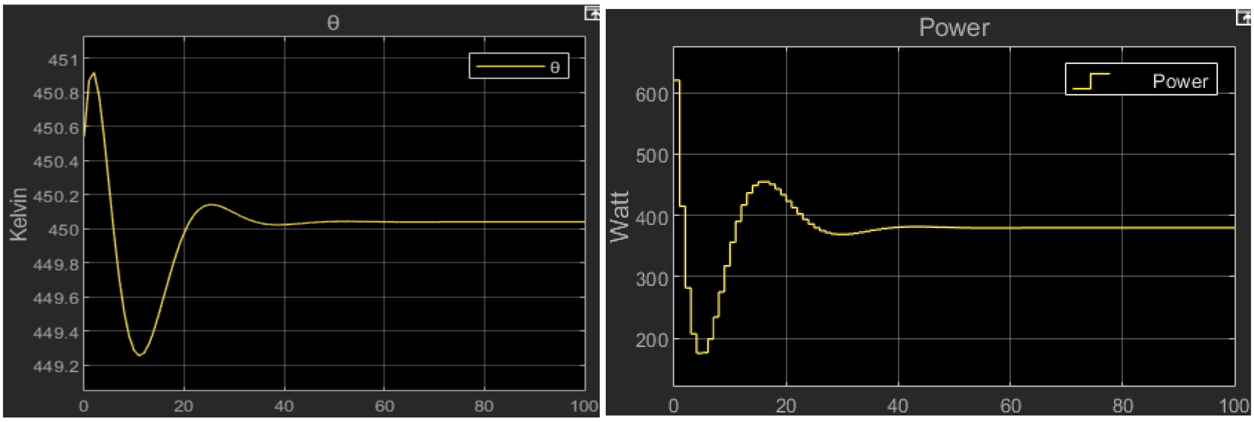

Fig 4.4: Theta variable and Power of the burner

We can appreciate that starting from an initial condition equal to:

$$x_0 = \begin{bmatrix} 897 \\ 0.2 \\ 450.54 \end{bmatrix}$$

That is to say:

- 3 [m] under the target (900 meters of altitude).
- 0.2 [m/s] of ascending vertical velocity.
- 450.54 [Kelvin] in the envelope of the Balloon.

Thanks to the action of the burner (input signal) the system reached the final state:

$$x = \begin{bmatrix} 900 \\ 0 \\ 450.04 \end{bmatrix}$$

That is equal to the equilibrium point chosen. Thus, the stabilization is completed.

5. STEP REFERENCE TRACK

Once we have guaranteed the stabilization of the equilibrium point, we can now focus on the second important problem of the task that is the tracking. In this section we want to make the system capable to track asymptotically a reference signal, in this case a step. To solve the problem let's make use of the integrator strategy. It is based on the following system:

$$\begin{cases} x_{k+1} = Ax_k + Bu_k + B_d d_k \\ y_k = Cx_k \\ q_{k+1} = q_k + y_k - r_k \end{cases} \qquad \textbf{(5.1)}$$

It represents the linearized-discretized system around the equilibrium point. Where:

- d_k and r_k are step signals representing respectively the amplitude of the disturbance and of the reference.

Third equation introduces a new state (the integrator) responsible for the storage of the tracking error. When the system reaches the equilibrium, we have:

$$\begin{cases} \bar{x} = A\bar{x} + B\bar{u} + B_d\bar{d} \\ \bar{y} = C\bar{x} \\ \bar{q} = \bar{q} + \bar{y} - \bar{r} \Rightarrow \bar{y} = \bar{r} \end{cases} \qquad \textbf{(5.2)}$$

So, the output $\bar{y}$ converges to $\bar{r}$ ensuring no tracking error and perfect rejection of the disturbance $\bar{d}$ acting on the system.

Let's implement this strategy for our system. The Matlab file responsible for the step track problem is ***Balloon_Step_Track.mlx*** and here the main code lines:

Define the equilibrium:

```
x1eq=900; %personal choice
x2eq=0; %trivial
x3eq=(V*rhoa*theta_a)/((V*rhoa)-m); %according to the equation of the equilibrium for x3
xeq=[x1eq;x2eq;x3eq];
ueq=k*(x3eq-theta_a); %according to the equation of the equilibrium for u
```

Compute the linearized model around [xeq,ueq]:

```
[Ac,Bc,Cc,Dc]=linmod("Balloon_model",xeq,ueq);
```

Discretize the linearized model in order to apply the LQ strategy:

```
Balloon_linearized_Tc=ss(Ac,Bc,Cc,Dc);
Ts=1; %sampling Time
Balloon_linearized_Td=c2d(Balloon_linearized_Tc,Ts);
[Ad,Bd,Cd,Dd]=ssdata(Balloon_linearized_Td);
```

Let's define the augmented system:

```
[ny nx]=size(Cd);
[nx,nu]=size(Bd);
A_aug=[Ad zeros(nx,ny);
       Cd eye(ny)];
B_aug=[Bd;
       zeros(ny,nu)];
C_aug=eye(nx+ny);
D_aug=zeros(nx+ny,nu);
Balloon_augmented=ss(A_aug,B_aug,C_aug,D_aug,Ts);
```

Define the parameters for the LQ strategy (tracking problem case):

```
Cz=[0 0 0 1;
    0 0 0 0];
rho=5;
Dzu=[0;
    rho]; %we have just 1 actuator (the burner of the Balloon)
Q=Cz'*Cz;
R=Dzu'*Dzu;
M=Dzu'*Cz;
```

Compute the optimal gains:

```
[F_inf,P_inf,eig_inf]=lqr(Balloon_augmented,Q,R,M');
```

Extract the gain related to the state and to the integrator:

```
F=F_inf(1,1:nx);
Fi=F_inf(1,nx+1:end);
```

Define the reference (step):

```
ref_altitude=400;
```

Let’s give some explanations:

1. The equilibrium point is the same of the previous case.
2. LQ strategy in the tracking problem case operates on an augmented discrete linear system composed by the system plus the state of the integrator. So, as before, the model has been linearized and discretized on the equilibrium. At the end, system *(5.1)* without considering the disturbance was composed.
3. After the optimization, the optimal gain for the state feedback can be split into two contributes:

$$-F * x = -[F_1\ F_2\ F_3 | F_i] \begin{bmatrix} z \\ \dot{z} \\ \theta \\ q \end{bmatrix}$$

- $[F_1\ F_2\ F_3]$ represent the components of the gain acting on the states.
- F_i represents the feedforward contribution acting on the tracking error.

The on-line part is implemented in the Simulink file ***Balloon_Step_Track.slx*** and here a snapshot of it:

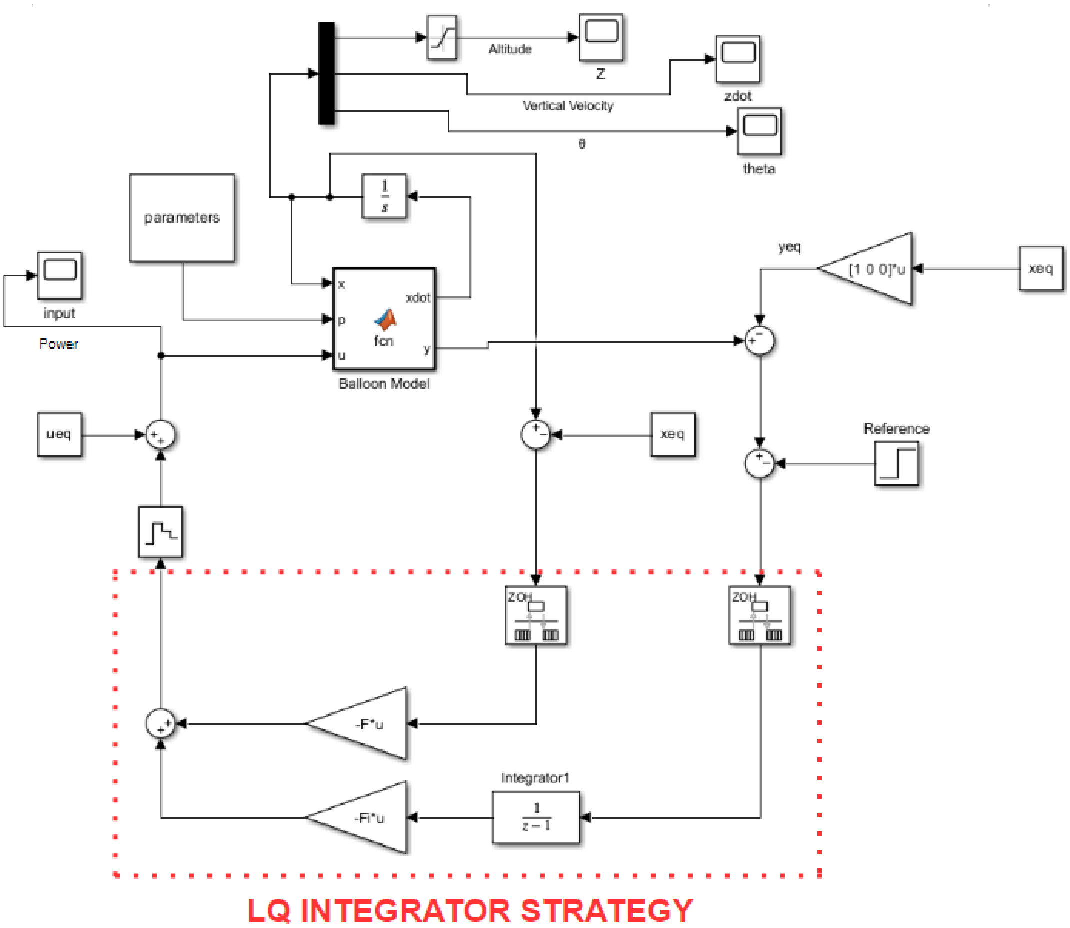

Fig. 5.1: LQ Integrator Regulation (Simulink)

After removing the bias on the state and on the output (we are interested in regulating the altitude, so the first state component), signals are sampled in order to apply the discrete linear strategy. Input signal is obtained by summing two contributions:

1. The state feedback term.
2. The feedforward term obtained using an integrator block acting on the error signal (difference between the output and the reference).

The results of the simulation are shown in the following figures:

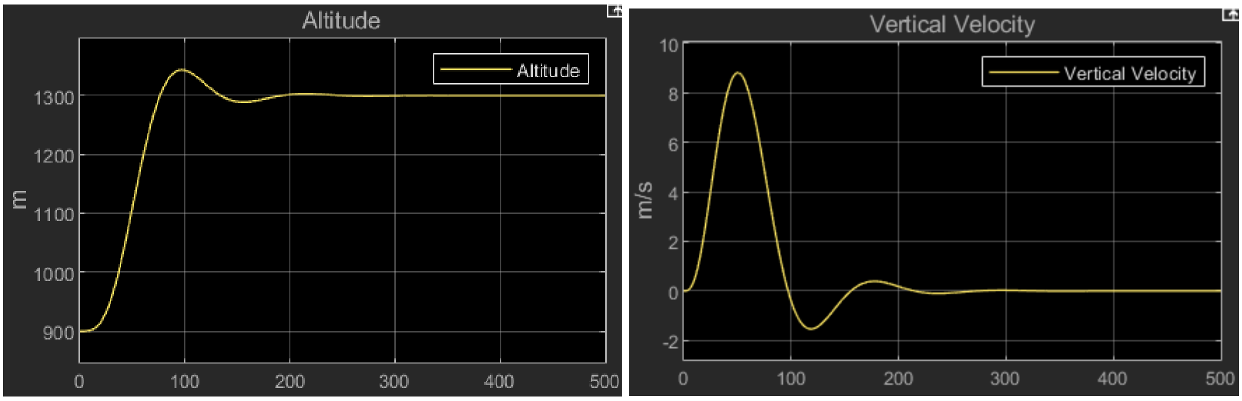

Fig. 5.2: Altitude and Vertical Velocity

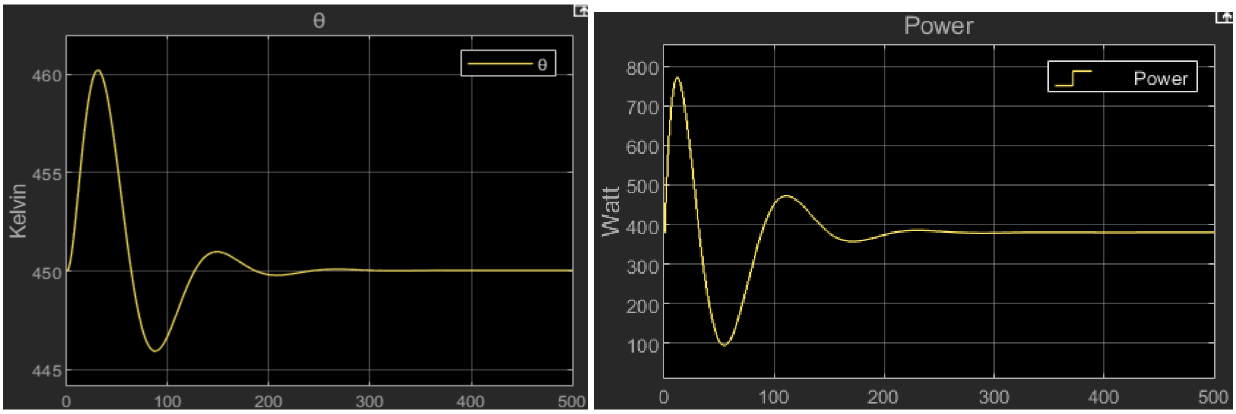

Fig. 5.3: Theta variable and Power of the burner

In the code we have considered a reference amplitude of 400 [m]. This means that the balloon should operate the following manoeuvre: starting from the equilibrium (suspended in the air at 900 [m] of altitude with no vertical velocity) it increases its altitude up to a quota of 900 [m]+400 [m]=1300 [m] maintaining this level. We can appreciate from the results that it effectively happens in about 300 [sec] (5 minutes), a reasonable timing considering factors such as the comfort of the passengers. At the end of the manoeuvre the balloon reaches a new equilibrium point equal to the previous one except for the altitude.

Now we want to investigate the behaviour of the balloon in a particular scenario. So far, we succeed to reach the target in terms of altitude but what could happen if suddenly a gust of wind of constant intensity hits the balloon? To analyse this case let's introduce the disturbance d_k discussed previously and modelled as a step because of its constant amplitude. The Simulink model becomes:

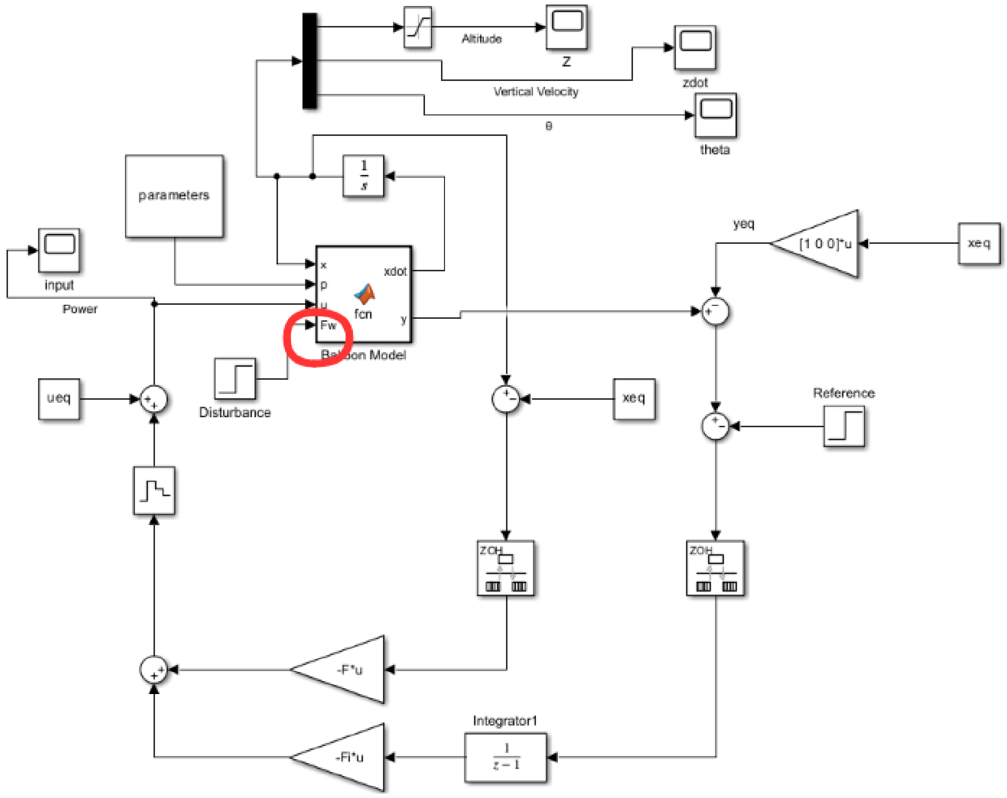

Fig. 5.4: LQ Integrator Regulation with disturbance (Simulink)

In the model the disturbance appears as a new input F_W for the system. Its internal evolution changes in order to take into account this new signal in the following way:

$$\begin{cases} \dfrac{dx_1}{dt} = x_2 \\ \dfrac{dx_2}{dt} = \dfrac{F_W}{m} - \dfrac{b}{m}x_2 - g + \dfrac{V}{m}g\rho_a\left(1 - \dfrac{\theta_a}{x_3}\right) \\ \dfrac{dx_3}{dt} = \dfrac{k}{c}\theta_a - \dfrac{k}{c}x_3 + \dfrac{u}{c} \\ y = x_1 \\ x(0) = x_0 \end{cases}$$

So, it affects the vertical velocity and, hence, the altitude. In particular, let's approximate the balloon to a mass point. The scenario of the analysis is the following:

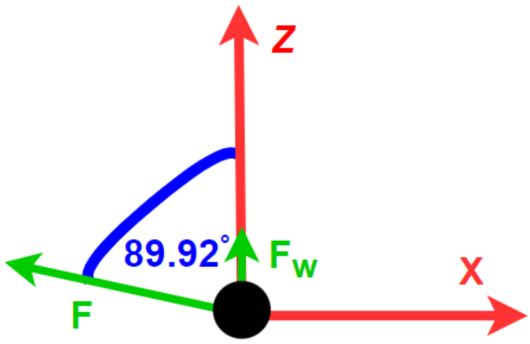

Fig. 5.5: Scheme of the gust of wind

Where:

- F is the force due to the gust of wind and it is equal to 215165 [N] (it models a wind velocity around 35 [km/h]) computed according to the formula of the drag resistance:

$$F = \frac{1}{2}\rho_a bAV^2$$

With A equal to the section of the balloon exposed to the wind velocity V.

- F_w is the vertical component of the wind force and according to the trigonometric formula is equal to 300 [N].

Let's analyse the results to check if the strategy was effectively capable of rejecting the disturbance. Without modifying the LQ parameters and considering a step time for the disturbance of 280 [sec], the outcome of the simulation is represented in the following figures:

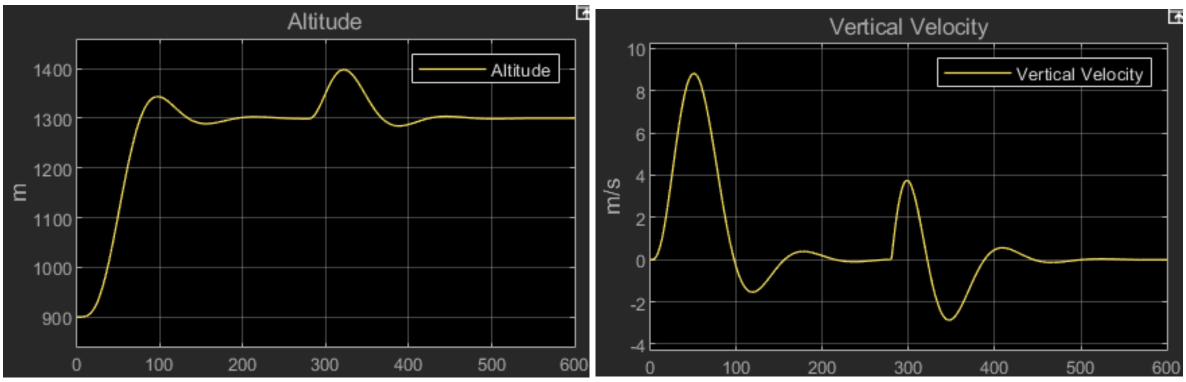

Fig. 5.6: Altitude and Vertical Velocity

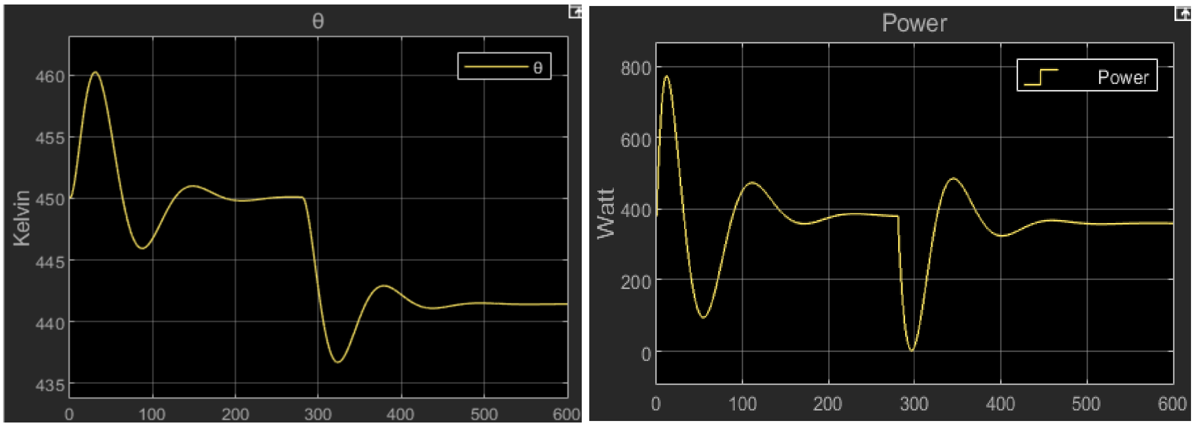

Fig. 5.7: Theta variable and Power of the burner

We can state that the balloon was capable of perfectly rejecting the sudden and constant disturbance of the wind.

6. SINUSOIDAL REFERENCE TRACK

In this final section we want to focus on a different tracking problem regarding a sinusoidal reference. To sort off this problem, it is sufficient to implement the same control logic of the previous case. The Matlab file ***Balloon_Sinusoidal_Track.mlx*** is responsible for defining all the parameters of the integrator LQ strategy. It is almost equal to the step track case except for the tuning of the rho parameter and of the reference amplitude:

```
rho=0.08;
ref_altitude=4;
```

The parameter rho has been decreased in order to obtain a stronger actuator force in charge to track the sinusoidal. The manoeuvre of the balloon should be the following: starting from the equilibrium (suspended in the air at 900 [m] of altitude with no vertical velocity) it modifies its altitude by means of a sinusoidal trajectory, so a periodic evolution with a maximum value of 900 [m] + 4 [m] = 904 [m] and a minimum value of 900 [m] - 4 [m] = 896 [m]. The Simulink file ***Balloon_Sinusoidal_Track.slx*** operates the on line-part:

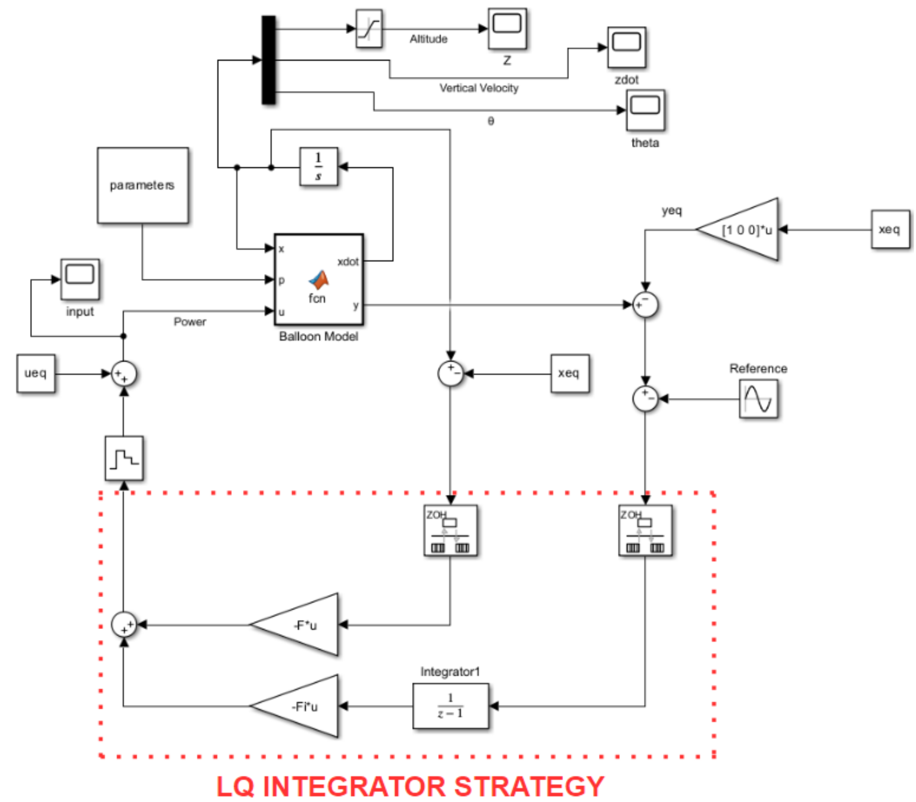

Fig. 6.1: LQ Integrator Regulation (Simulink)

It is very similar to the step case exception for the nature of the reference.

The results are the following:

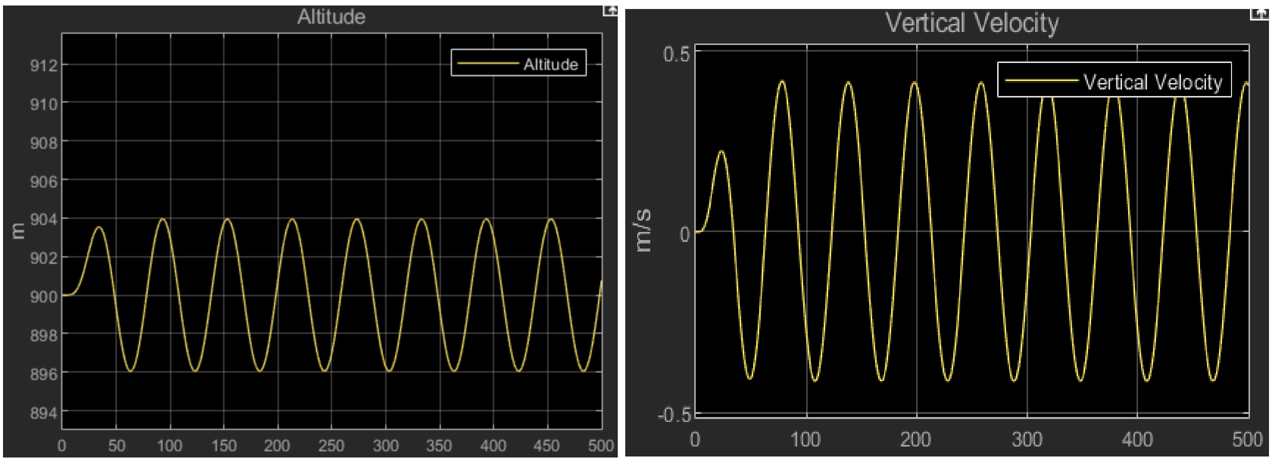

Fig. 6.2: Altitude and Vertical Velocity

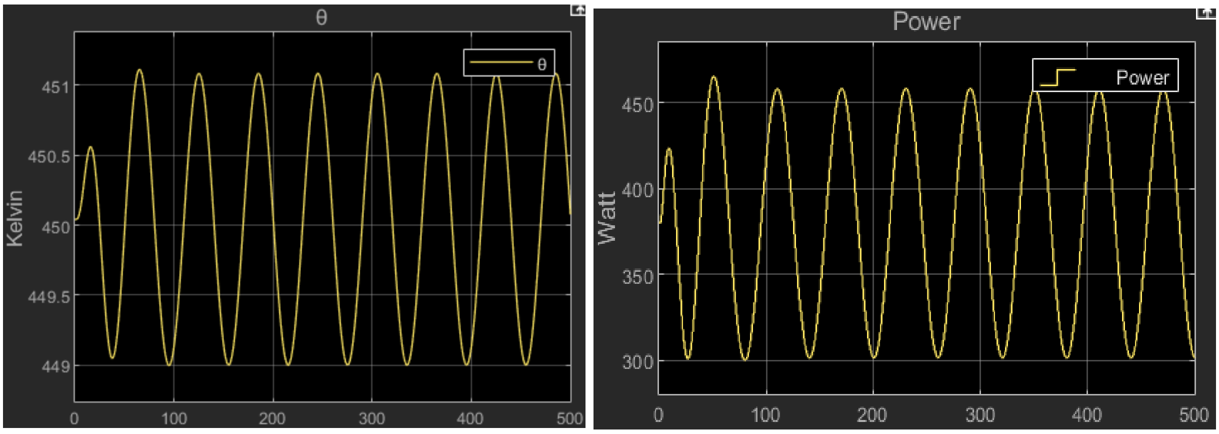

Fig. 6.3: Theta variable and Power of the burner

We can appreciate that the balloon succeeded to follow the desired trajectory (probably a bad experience for the passengers).

To complete the analysis let's introduce the disturbance under the same hypothesis described before.

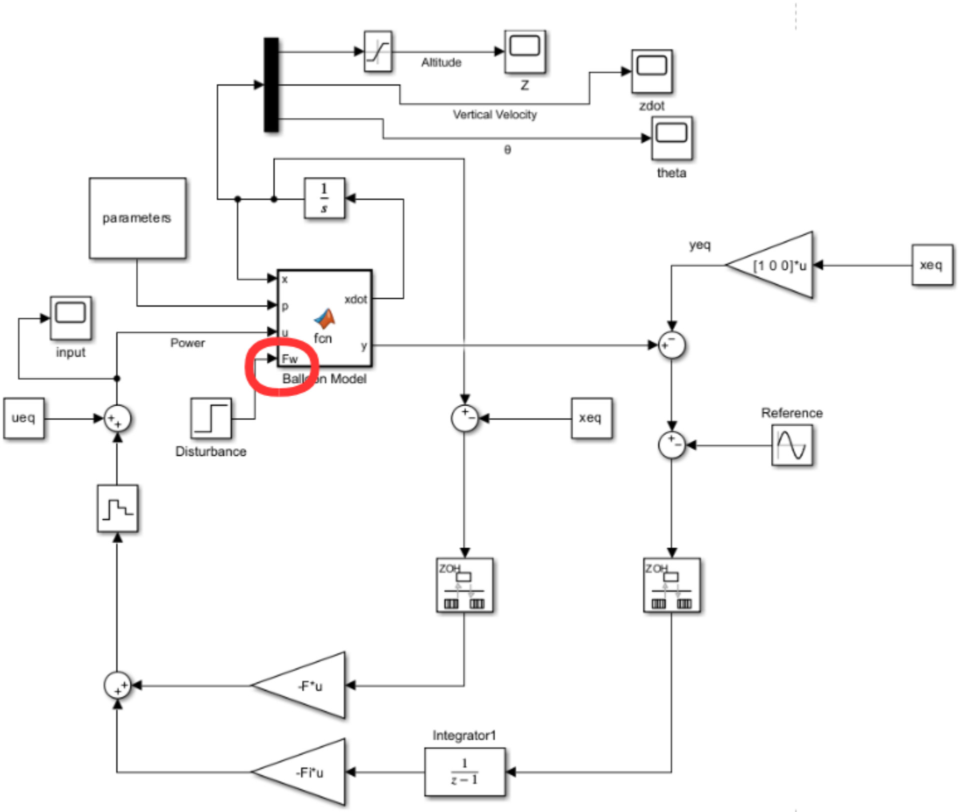

Fig. 6.4: LQ Integrator Regulation with disturbance (Simulink)

It provides the results shown in the figures:

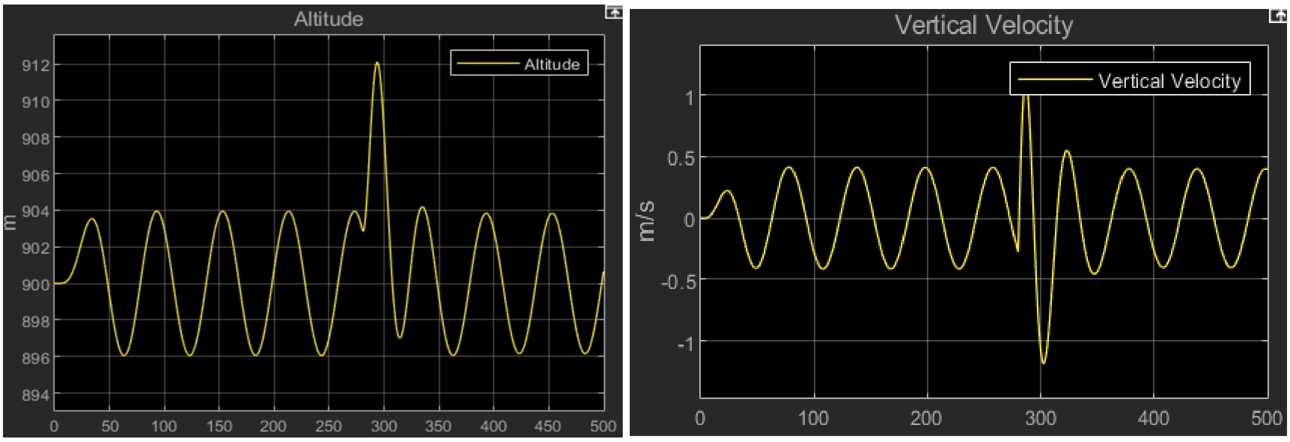

Fig. 6.5: Altitude and Vertical Velocity

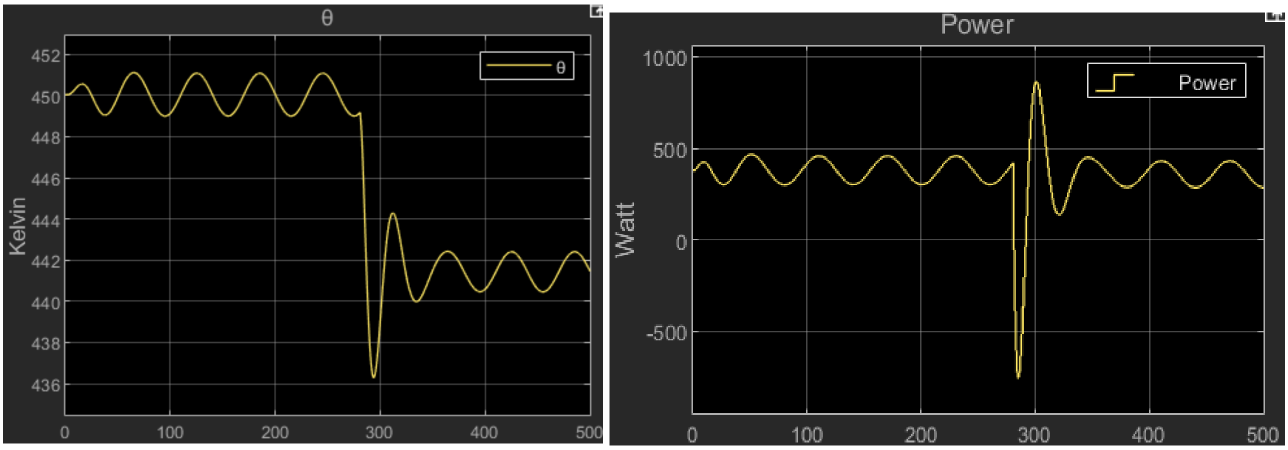

Fig. 6.6: Theta variable and Power of the burner

Thus, we can state that also in the case of a sinusoidal reference the balloon guaranteed perfect rejection of the disturbance.

However, further explanations are necessary. During all the simulations in this work we have obtained realistic values regarding the state variables and the actuator burner. In this case we can put our attention on the fact that once the disturbance arises there is sudden variation of the temperature inside the envelope and, especially, of the power. Moreover, in the time trend of the power there is a time interval in which it is negative. This is due to the solution of the differential equations and, hence, it comes out from numerical reasons. The problem is that a negative power has not a physical meaning since the burner is only capable to supply energy to the envelope instead of absorbing from it. However, the introduction of a saturation block set to a minimum value of zero generates instability in the system which, to reject the disturbance, needs the negative values shown above. A possible solution could be the use of an exhaust valve. It should be always closed in order to maintain the temperature in the envelope, but in some particular cases when it is required to have fast variations of the temperature, probably due to sudden disturbances and associated to negative values of the power, opening the valve we could obtain the variations put in evidence in the simulation without violating physical constraints.